

INDIAN ECONOMY

**LEVEL
II**



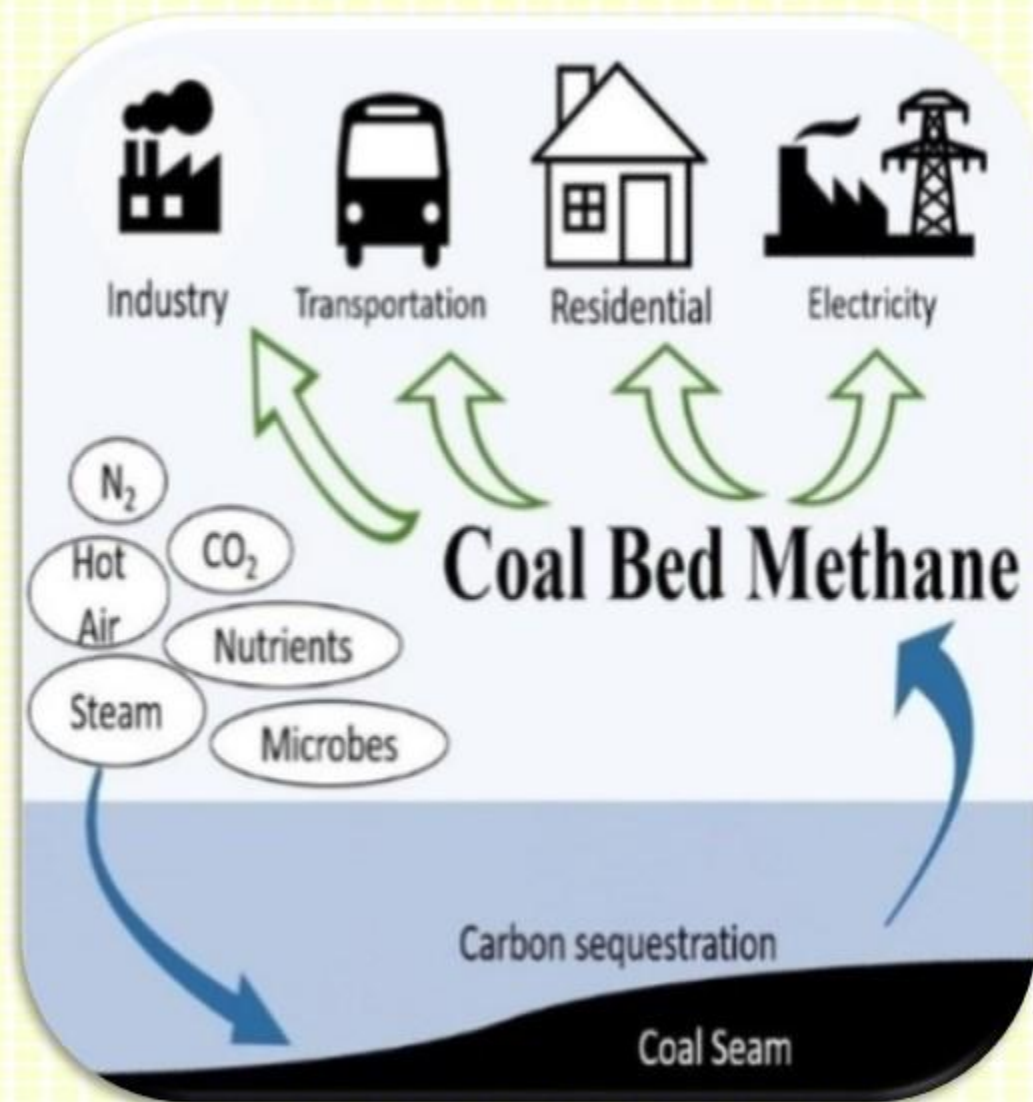
LECTURES

**ENERGY, MINES & MINERALS ...
AN OVERVIEW OF THE RECENT
DEVELOPMENTS - PART - II**

MODULE - 44

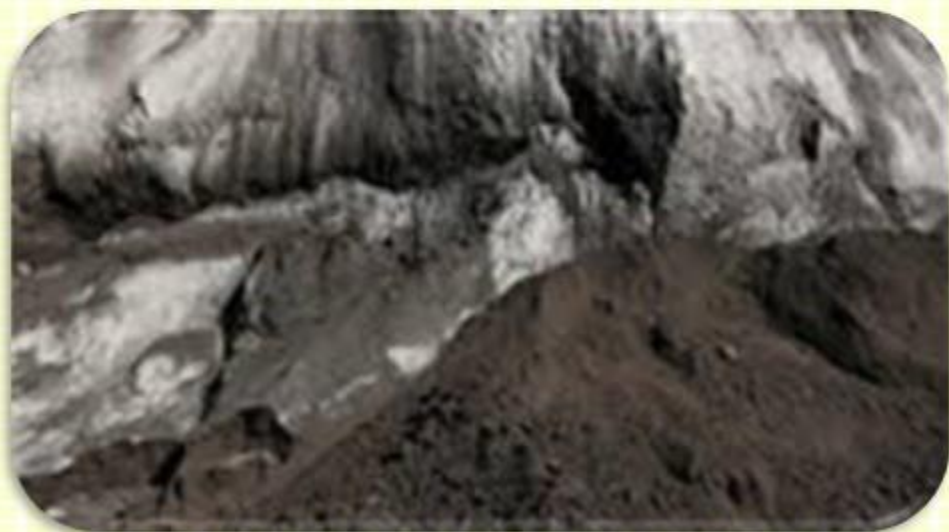


IT IS 25 YEARS FOR COAL BED METHANE (CBM) POLICY!



- The Government had formulated a CBM Policy in 1997.
- The Ministry of Petroleum and Natural Gas became the administrative ministry.
- Directorate General of Hydrocarbons (DGH) was made the nodal agency for development of CBM.
- Both Petroleum and Coal Ministries were jointly made responsible for developing the resource.
- Delineation of CBM blocks was to be done only in consultation with the Coal Ministry.
- During 2000-2008, a total of 33 CBM blocks were awarded to different national and private oil and gas companies for exploitation.

IT IS 25 YEARS FOR COAL BED METHANE (CBM) POLICY!



BUT, COULD NOT ACHIEVE SUCCESS!

- Only 1.87 mmscmd of CBM production could be achieved till March 2019. It was only from four operational blocks — two in Raniganj and one each in Sohagpur and Jharia.
- 20 CBM blocks are apparently proposed to be relinquished. The remaining blocks are still in different stages of development.
- Even a demonstration project taken up in 1999 could achieve only some of the envisaged objectives.
- Development of CBM commenced in Australia almost during the same period as that of India.
- Australia CBM development reached an impressive 109 mmscmd, and India's was less than 2% of the level achieved by Australia.

WHAT ARE THE POSITIVE DEVELOPMENTS IN RECENT TIMES?

- There have been two major reforms between 2015 and 2018.
- Public sector coal companies were permitted to explore and extract CBM from the leasehold areas held by them without seeking separate lease for gas extraction.
- The second aspect is - freedom to fix the price and sale of gas was extended by the Government to all operators of CBM.



WHAT SHOULD BE DONE TO MAKE CBM POLICY A SUCCESS?

- We urgently require enabling mechanism in terms of policy initiative and administrative transparency, coupled with requisite data / information depository.
- It requires international collaboration. International technology providers/operators may select an Indian partner of their choice to implement CBM projects in India. This will not only ensure introduction of contemporary technology to suit Indian conditions but also mitigate risk of investment.
- Technology transfer, however, should be mandatory as per a clearly-laid-out roadmap in sync with 'Atmanirbhar Bharat' objectives.

It is time to revisit the policy to invigorate Methane extraction.



GROUND-MOUNTED SOLAR SYSTEMS



ROOFTOP SOLAR SYSTEMS



- India is focusing primarily on large scale ground-mounted solar photovoltaics. But it failed to exploit several benefits of Decentralised Renewable Energy (DRE) options, which include reduction in Transmission and Distribution (T&D) losses.
- One of the primary benefits of solar PV technology is that it can be installed at the point of consumption, so that capital-intensive transmission infrastructure can be reduced.

ROOF TOP SOLAR (RTS) ... AN ABYSMAL FAILURE!

5 There is limited financing for residential consumers and Small and Medium Enterprises (SMEs), who want to install RTS.

4 The subsidised tariffs charged to residential customers undermine the economic viability of installing rooftop solar panels.



6 Lack of awareness.

1 The efficiency of the solar panels vary on any given day depending on how bright the sunlight is and the solar panels also produce no electricity during the night. Electricity is required the most during night time and even with solar, grid connectivity is required. To avoid this problem, it requires battery storage. The battery storage is too costly. Moreover, the storage technology is still evolving.

3 It puts restrictions on the usage for other purposes.

2 Lack of interest by DISCOMs to purchase power through net-metering.

ROOF TOP
SOLAR IS
DRAGGING
THE OVERALL
PROGRESS
BADLY!

HOW DID THE TRANSITIONAL SHIFT FROM FOCUS ON WIND ENERGY TOWARDS SOLAR ENERGY OCCUR IN INDIA?

- Wind power generation blossomed on the foundations of the Electricity Act, 2003 and the establishment of a robust home-grown manufacturing base.
- The wind sector gained momentum, thanks to private investments and government initiatives coupled with regulatory support.
- But things changed when Feed-In-Tariff (FIT) was replaced with the reverse bidding mechanism and the advent of solar into the renewable energy mix.

- Backed by global supply of solar cells and modules and favourable policies, solar power emerged more competitive than wind power in recent times and the wind power capacity addition started to plummet.



WHY SOME EXPERTS CONSIDER WIND POWER BETTER THAN SOLAR POWER?



Wind helps balance the grid, which in turn enables higher penetration of solar!

- Experts feel, from a technological and resource perspective as well as from the commercial angle, wind is a better alternative than solar.
- The reasons why wind is more desirable include
 - ✓ Higher capacity utilisation, and generation (solar cannot produce throughout the day).
 - ✓ Potential of complementing solar, providing a more consistent and viable generation profile.
 - ✓ Foothold into the manufacturing ecosystem, not like solar, where solar modules/cells are mostly imported.

HAVE A GLANCE AT THE POTENTIAL OF WIND ENERGY!



Offshore is still at the tender stage, whereas the onshore wind capacity is at around 40-45 GW by the end of 2022.

- The National Institute of Wind Energy estimates a total wind energy potential of 302 GW at a hub height of 100 metres above ground, which more than doubles to over 695 GW at a hub height of 120 metres.
- In addition, India has a potential of 174 GW of offshore wind resources, as per an estimate by the Global Wind Energy Council.
- The offshore is primarily concentrated in Gujarat and Tamil Nadu coasts, whereas onshore wind is also highly concentrated in Tamil Nadu, Gujarat, as well as in Maharashtra and other states.



ORIGINAL EQUIPMENT MANUFACTURERS IN THE WIND SECTOR GONE DOWN SUBSTANTIALLY!

- India was a global wind manufacturing hub, with more than 80 per cent of the components sourced locally, and with full supply chains backed by highly skilled manpower to operate this sector.
- Today, in spite of this head-start and the potential for strong export opportunity, the sector is going through a rough patch.
- Around 5 to 6 years ago, India had around 30 Original Equipment Manufacturers (OEMs) in the wind sector. The number has dwindled to around a dozen now.

FINALLY, WHAT ARE THE REASONS FOR THE DOWNFALL OF THE WIND POWER SECTOR?



OVERALL, THE
APATHY OF THE
GOVERNMENT IS
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5 For cost-effective development of wind plants, incidental issues like land procurement and coordinated development of projects, timely construction of transmission facilities, have become the bottlenecks quite often.

4 Lack of recognition of true cost of renewables by making subsidies explicit. To expand further, wind and solar cannot have the same tariff rates.

1 Lack of interest shown by the Discoms, along with risks of payment delays.

2 Switch from Feed-In-Tariff (FIT) to reverse auctions before the industry was actually ready for it.

3 MSME sector, which actually contributed to growth in installed capacity for captive use, has been affected with the bid size raised to a minimum of 50 MW.

FINALLY, WHAT ARE THE REASONS FOR THE DOWNFALL OF THE WIND POWER SECTOR?

8 Getting financing for renewable projects is really challenging and high cost of debt financing, short repayment tenures, and over-reliance on banking institutions compounded the problems. The sector needs novel financing models.



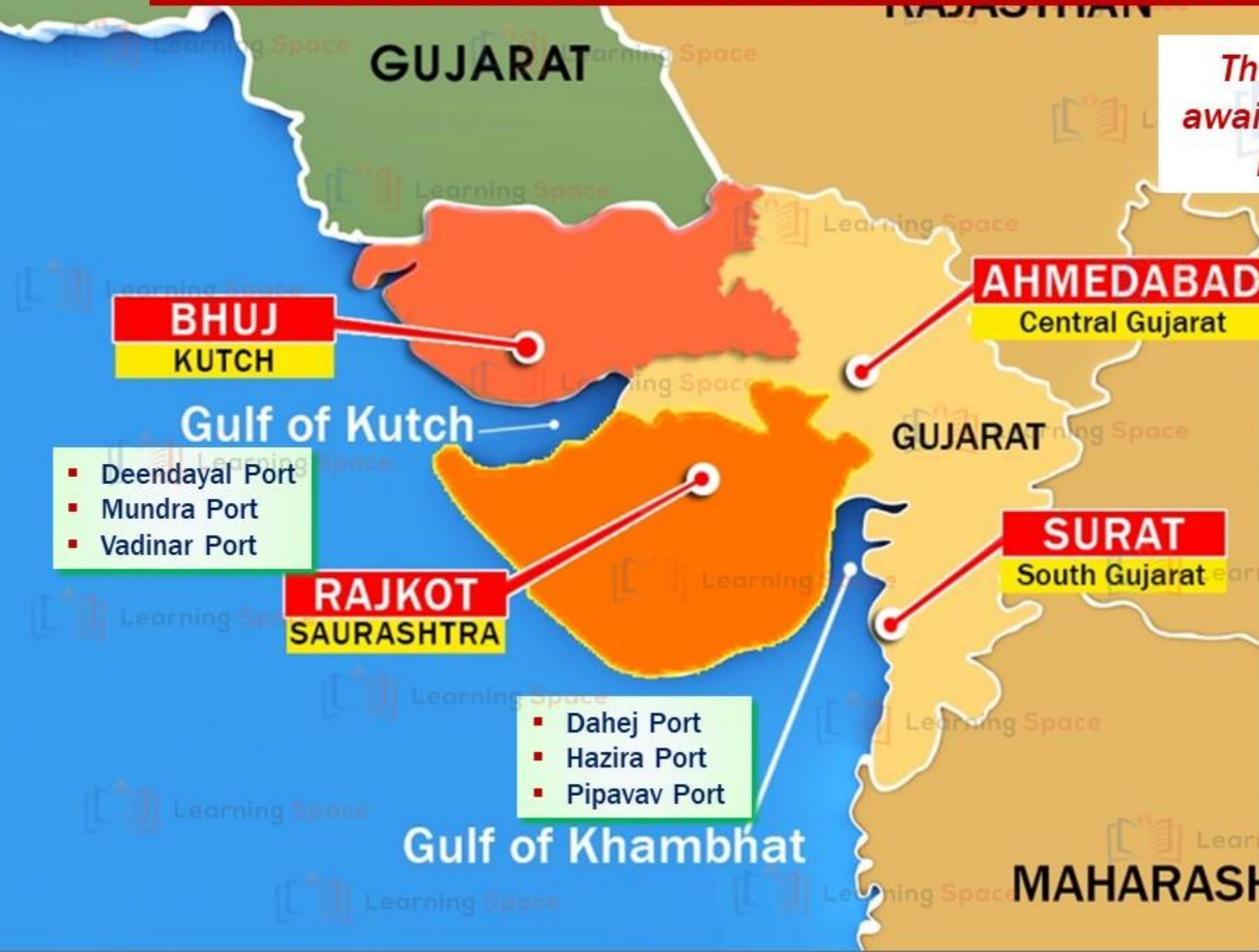
6 Lengthy processes in land acquisitions and payment of compensation and environmental concerns resulted in time and cost overruns.

7 In the areas where poor infrastructure for access to wind turbine generators was in existence, extortions and theft often plagued smooth operations.

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KUTCH ... FROM FOSSIL FUELS TOWARDS RENEWABLE ENERGY!



The water starved district still awaits a full-fledged access to the Narmada Canal Network.

- Kutch district is spread over 45,600 sq.km, a little bigger than Switzerland.
- During the past two decades, Coal based power plants came up in Kutch.
- Tata Power's 4,000 MW thermal plant under the Central Govt's Ultra Mega Power Project (UMPP) scheme and Adani group's 3,300 MW Super-critical Mundra Power Project turned Kutch into a power house.

WORLD'S LARGEST SINGLE LOCATION HYBRID ENERGY PARK IS ALSO PLANNED IN KUTCH!



- The Prime Minister in the year 2022, laid the foundation stone for what is being dubbed as the World's largest single location hybrid energy park in Vighakot.
- It will be a 30 Giga Watt renewable energy hybrid park – Wind and Solar.
- It is considered a big step for India in its commitment towards climate change.
- The State Government estimates the project will get investments of Rs. 1.25 Lakh Cr. from Solar and Wind players, who will sell the electricity generated to the Solar Energy Corporation of India.

WHAT DO YOU UNDERSTAND BY GIGA FACTORIES?



- A Giga factory is a gigantic battery production facility that brings multiple companies and component makers together to scale up for cheaper battery manufacturing and the word was first coined by Tesla's CEO Elon Musk in 2013.
- The need for giga factories arises from huge demand for energy storage in the renewable energy sector and advanced batteries in the electric mobility segment.
- In 2022, the Power Minister stated that India is planning to float tenders for 4,000 Mega Watt Hour (4 GWh) battery storage and ancillary power generation.

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